

Roman engineering

The Romans built an empire that began with a few Italian states and spread to encompass most of Europe in the west, and to the Persian Empire in the east. They established their dominance in part through law-making, but also through developing advanced engineering techniques that enabled them to build straight roads irrespective of natural features, cities planned on rigid grids, aqueducts, viaducts, bridges, baths, arches and domes. Although some of their inventions and techniques simply improved on earlier (often Greek) ideas, their accomplishments surpassed any other civilization of the time.

Concrete made possible the construction of great vaults and domes, some of whose spans were not equalled until the development of steel in the 19th century. Amphitheatres allowed hundreds of spectators to see and hear spectacular displays, plays and special effects. Aqueducts, built from arches in one, two and three tiers, carried water in cement-lined channels, which ran along the upper level before diverting into pipes.



Roman aqueduct at Segovia, Spain